

Telecommunication Final Report



Faculty of Engineering

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Technology, Business and Design

Telecommunication Project 2023

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Task - 1

Plot the spectra for receiver 1 and one of the other four for all ranges for magnitude and phase (use e.g. pcolor). Maintain a proper frequency axis scaling... (also for the latter plots)

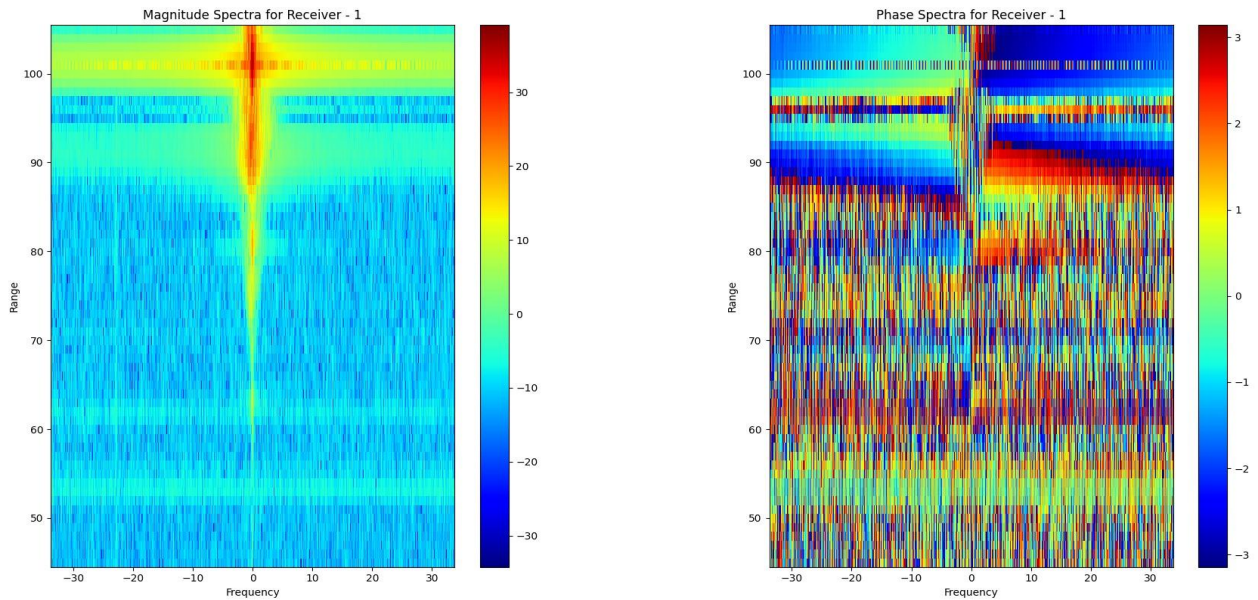


Fig 1.1 : Magnitude and Phase spectra for receiver channel - 1

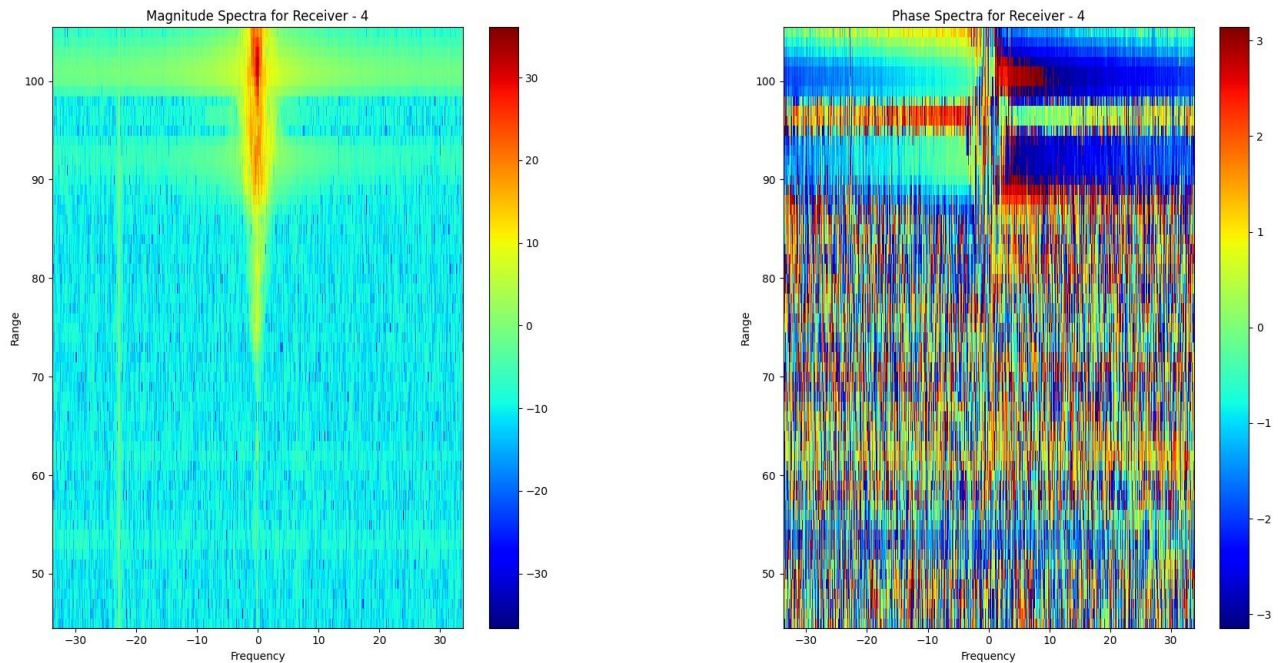


Fig 1.2 : Magnitude and Phase spectra for receiver channel - 4

Fig 1.1 and Fig 1.2 shows the obtained spectra plots of receiver 1 and receiver 4 for both magnitude and phase respectively. The range grid and frequency grid was properly scaled. The significance of applying the Fast Fourier Transform (FFT) to the raw data and plotting the spectra for magnitude and

phase of receiver channel lies in its capability to analyze radar signals by revealing their frequency content. The FFT allows us to transform the time-domain radar echoes into the frequency domain, enabling us to identify and examine the different frequencies present in the received echoes. This analysis provides valuable insights into the characteristics of the radar signals, helping us understand the underlying components and features that contribute to the overall signal. Based on the spectral visualization, it is to be observed that receiver 1 exhibits a lack of signal presence until approximately 60 km range. A minimal signal is noticeable from 60 to 85 km, while a significant increase in signal strength can be observed between 85 and 100 km. In contrast, receiver 4 starts to detect signals after approximately 70 km range. The signal strength gradually increases beyond the 88 km range, but the frequency range in which signals are present appears narrower compared to receiver 1.

Task - 2

Compare the spectra for:

- a) superposition of the time series of receiver channels 2-5
- b) combination of the individual spectra of receiver channels 2-5

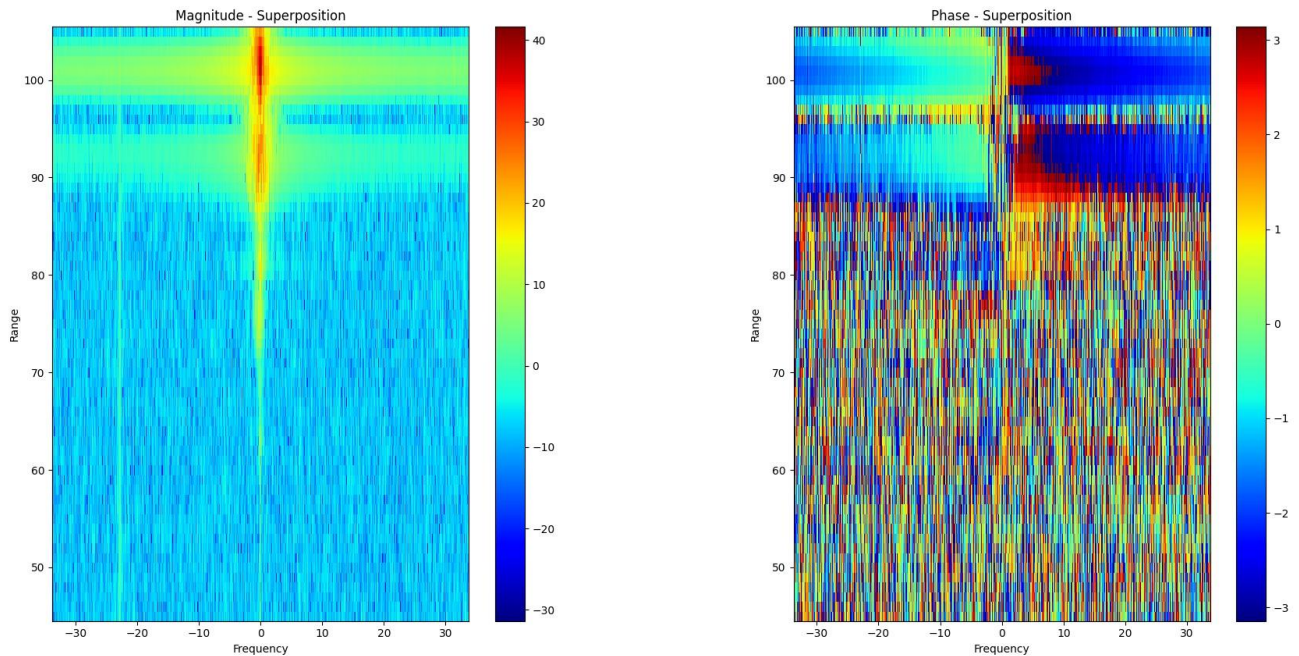


Fig 2.1 : Magnitude and Phase spectra for superposition of time series of receiver channel 2-5

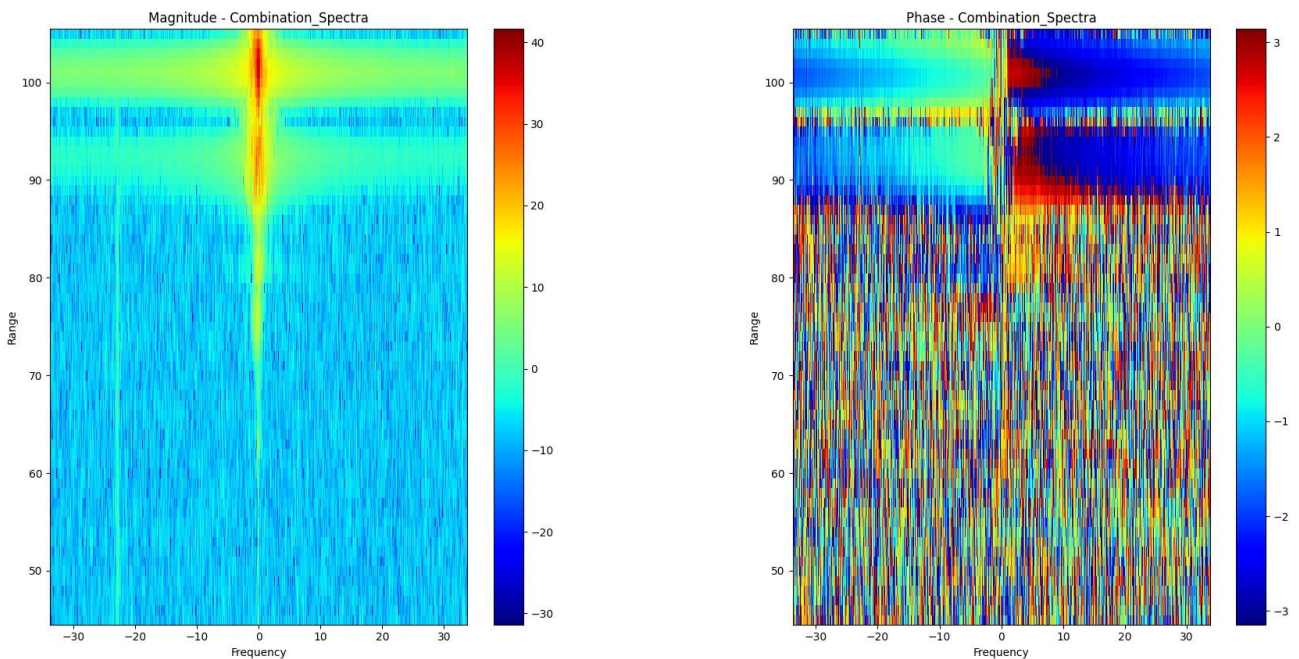


Fig 2.2 : Magnitude and Phase spectra for combination of individual spectra of receiver channel 2-5

Figures 2.1 and 2.2 depict the spectral representation of the combined signal from receivers 2 to 5 and the individual signals from each of these receivers, respectively. In the superposition of spectra for

receivers 2 to 5, the echo power from each receiver is summed together, resulting in a single combined signal. This combined signal represents the total power received by all five receivers at each time sample and range gate. Upon examining the spectral visualizations, it is evident that there is a minimal disparity between the plots for the superposition of receivers 2 to 5 and the combination of individual signals from these receivers. The observed signals are relatively weak until the range reaches approximately 72 kilometers, with only a slight indication of signal presence up to around 87 kilometers. Notably, a slightly stronger signal can be observed at a lower range, around 90 kilometers to 100 kilometers mark. Overall, the comparison between the two plots suggests that the combined power of receivers 2 to 5 does not significantly differ from the sum of the individual power spectra of these receivers.

Task - 3

Apply a suitable number of coherent and/or incoherent integration (CI/NCI) to the data and show the spectra for the receiver 1 and one of the other four for all ranges for magnitude and phase (use e.g. pcolor). Compare with 1 !

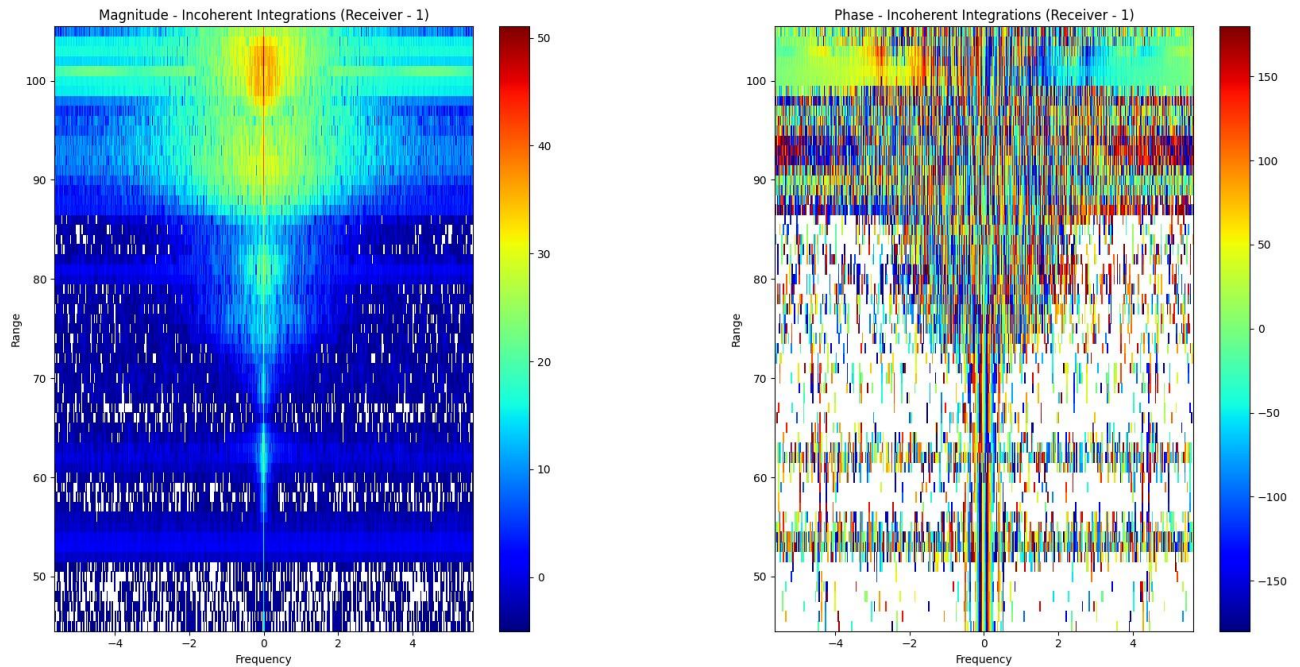


Fig 3.1 : Magnitude and Phase spectra for NIC of receiver - 1

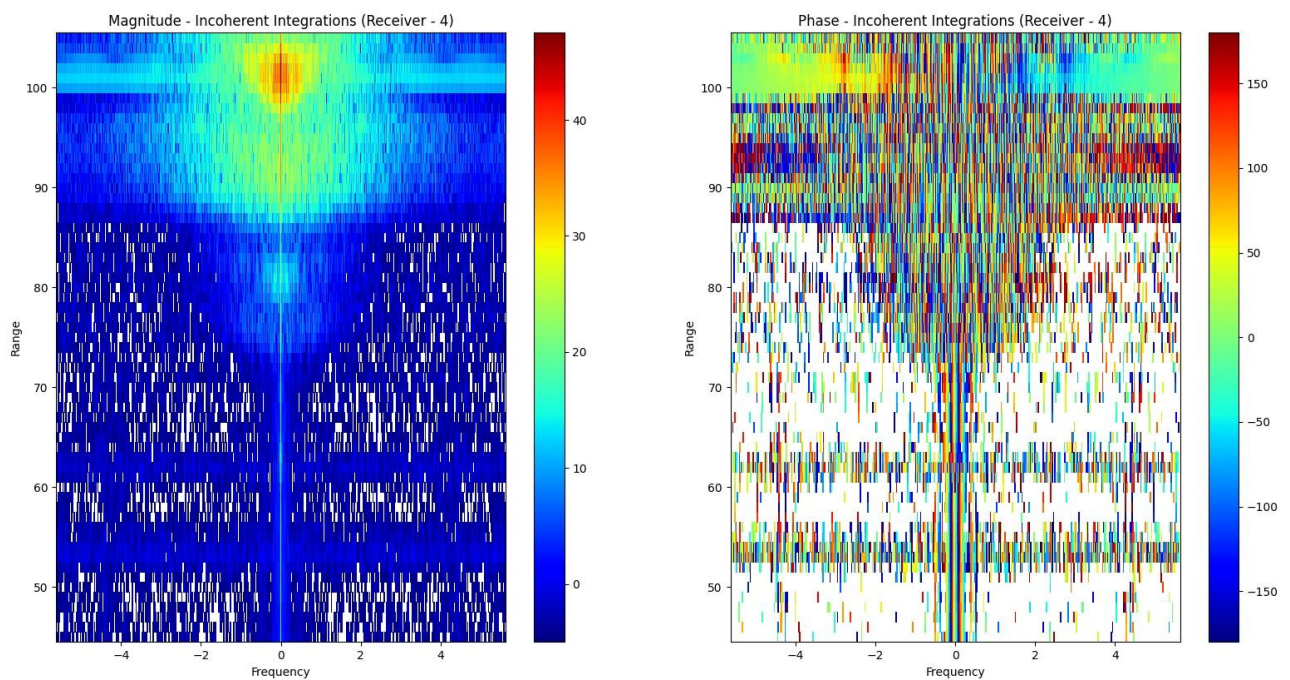


Fig 3.2 : Magnitude and Phase spectra for NIC of receiver - 4

Coherent integration and incoherent integration are techniques utilized in signal processing and communications to enhance the detection and estimation of weak signals in the presence of noise.

Coherent integration involves accumulating multiple signal samples coherently over a specified period. This technique aligns the phase of the received samples and combines them before further processing. By exploiting the phase relationship between consecutive samples, coherent integration enhances the signal-to-noise ratio. On the other hand, incoherent integration disregards the phase relationship between signal samples and focuses solely on their magnitudes or powers. It involves averaging the squared magnitudes of the received samples over a specific period. Incoherent integration reduces the impact of noise fluctuations and improves the signal-to-noise ratio. This technique is particularly useful when phase information is not critical.

Figure 3.1 and Figure 3.2 depict the plots of the magnitude and phase spectra, respectively, for the incoherent integration of receiver 1 and receiver 4. These plots reveal noticeable differences in the signal range between the two receivers. For receiver 1, the signal becomes noticeable around the 70-kilometer mark and increases thereafter, along with an expansion in the frequency range. In contrast, receiver 4 exhibits a lower signal level around 80 kilometers, and the signal starts to increase from approximately 88 kilometers. Additionally, the frequency range is relatively lower for receiver 4 compared to receiver 1. To enhance the visibility and interpretability of the integrated signal, SNR cleaning techniques can be applied.

Task - 4

Derive and show the cross-spectra for receiver channel 2-5 in magnitude and phase. Use the integrated data as CI/NCI applied in 3.

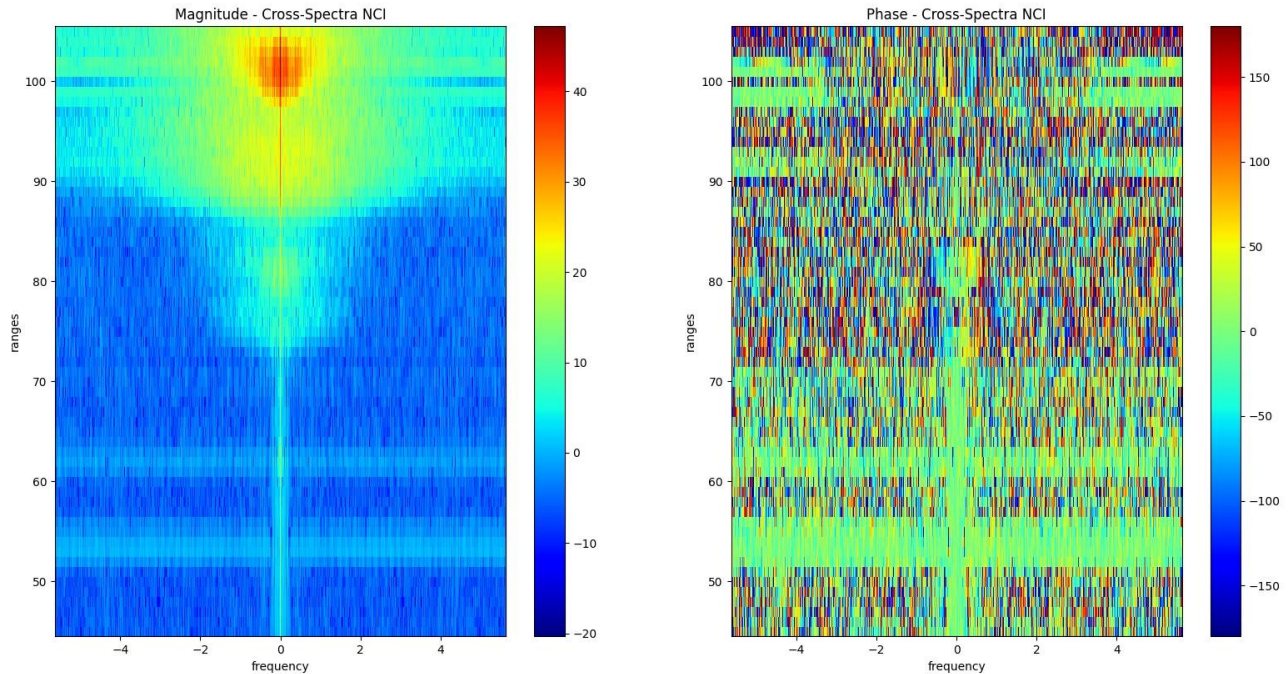


Fig 4.1 : Magnitude and Phase spectra for cross-spectra (NIC) of receiver 2-5

The cross-spectrum, also known as the cross-power spectral density, is a measure used in signal processing and spectral analysis to assess the relationship between two signals in the frequency domain. It provides insights into the coherence, phase, and magnitude between the signals at different frequencies. By examining the magnitude and phase of the cross-spectrum, one can determine the presence of correlated or coherent components between the signals at various frequencies. The magnitude of the cross-spectrum indicates the strength of the relationship, while the phase reveals information about the phase difference or time delay between the signals. In Figure 4.1, which displays the magnitude and phase of the cross-spectrum for the incoherent integration data, it can be observed that the signal strength is relatively low in the range of 75-85km. However, the signal strength increases notably beyond the 90km range. Additionally, the frequency range appears to be relatively low. It is important to note that SNR cleaning has not been applied in this analysis.

Task - 5

Get the 1st MAARSY RAW file and Apply a suitable number of coherent and/or incoherent integration to the data and show the spectra for the receiver 1 and one of the other three for all ranges for magnitude and phase (use e.g. pcolor).

Also derive and show the cross-spectra in magnitude and phase for receiver 2-4.

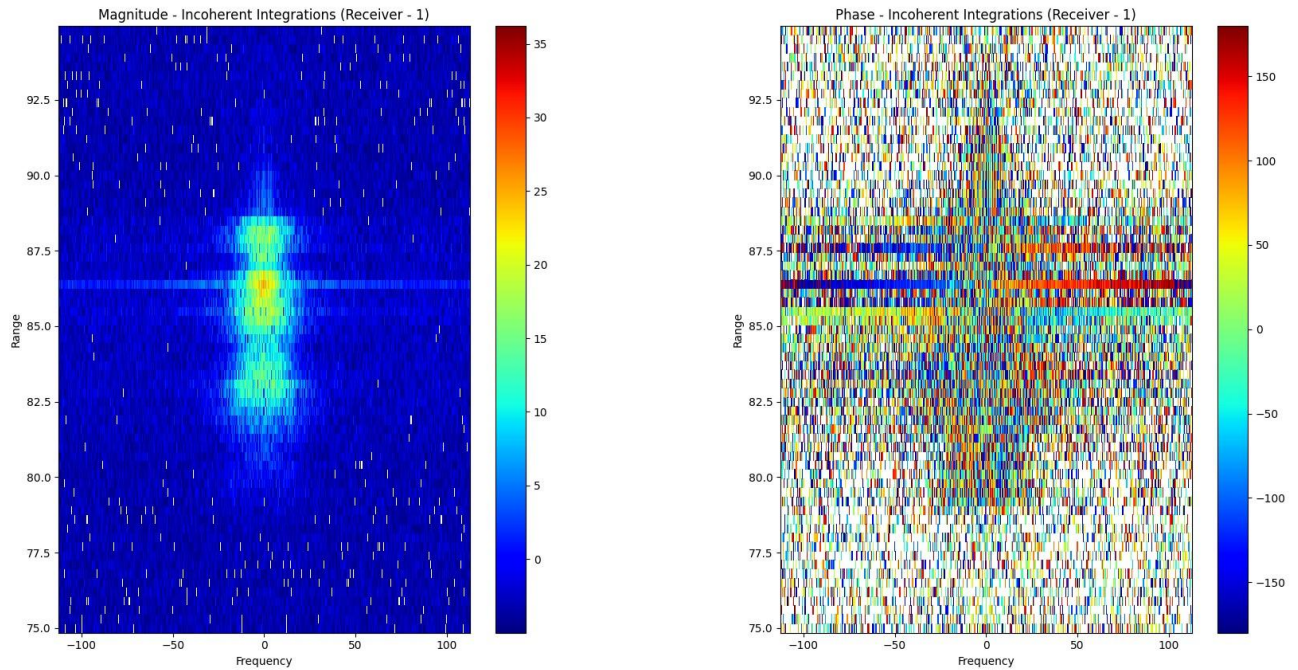


Fig 5.1 : Magnitude and Phase spectra for NIC of receiver - 1

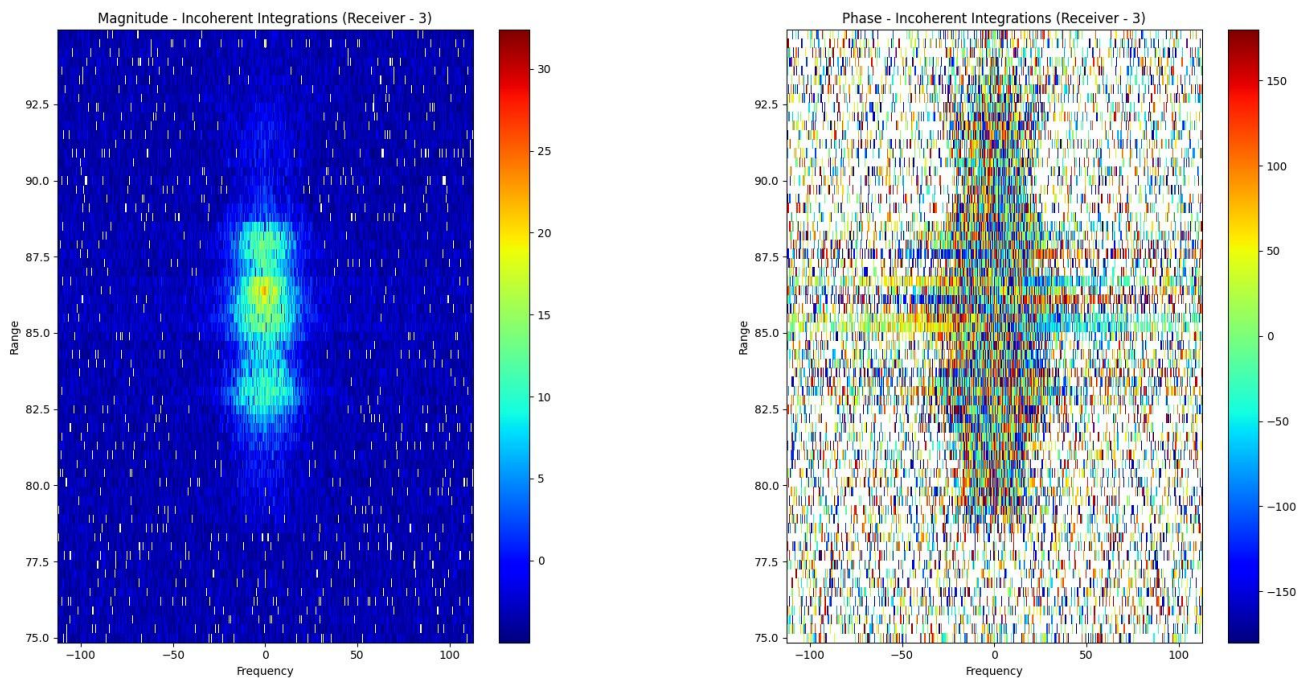


Fig 5.2 : Magnitude and Phase spectra for NIC of receiver - 3

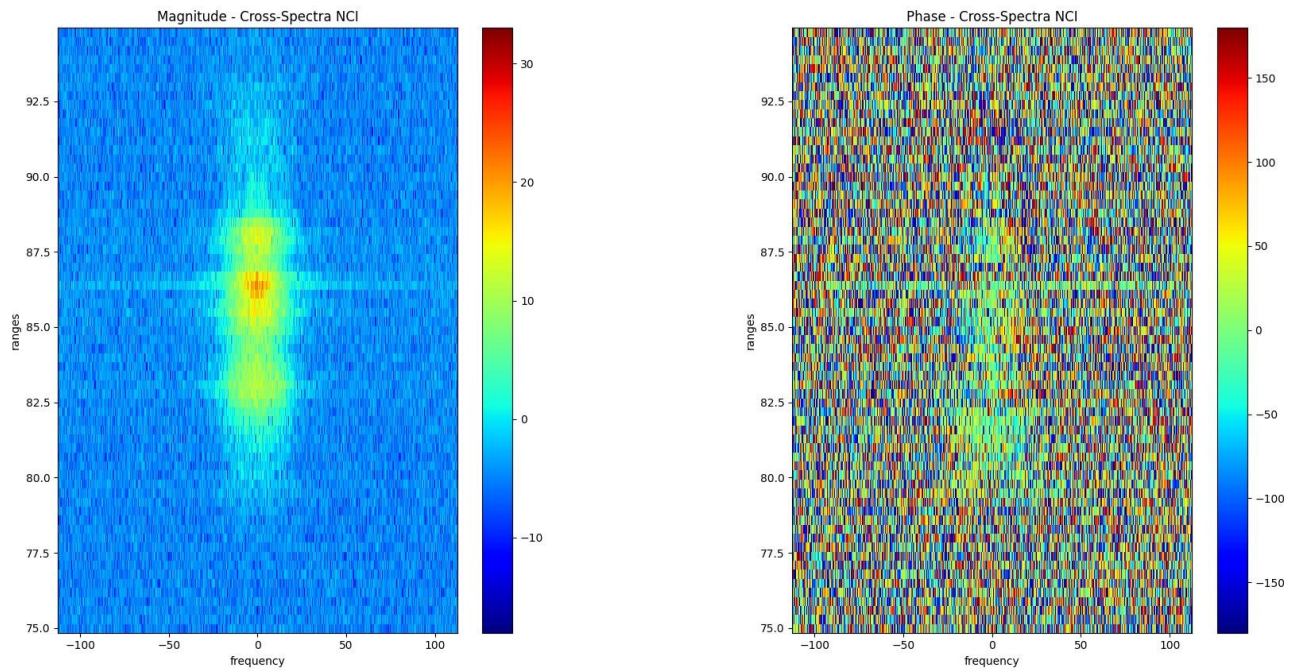


Fig 5.3 : Magnitude and Phase spectra for cross-spectra (NIC) of receiver 2-4

Figure 5.1 and Figure 5.2 depict the magnitude and spectrum plots resulting from the application of the incoherent integration technique to receiver 1 and receiver 3, respectively. Both cases involve the application of SNR cleaning. In receiver 1, no observable signal is present up to approximately 81 kilometers. However, a visible signal emerges between the 82 and 89-kilometer range, exhibiting a higher frequency range compared to other performed tasks. Receiver 3 follows a similar pattern, where a signal becomes apparent within the 82.5 to 88-kilometer range, albeit with a lower frequency range compared to receiver 1.

Figure 5.3 illustrates the magnitude and phase plot of the cross-spectrum derived from the incoherent integrated data. A distinct and robust signal is present around the 85-87 kilometer range, while the signal is dispersed across the range of approximately 80 to 90 kilometers. Notably, the frequency range is comparatively higher than that observed in Task 4.

Task - 6

Get the 2nd MAARSY RAW file and

- plot the echo power for all ranges for receiver 1 and the superposition of the other three.
- Derive the Angle-of-Arrival for the receiver channel 2-4 for each time sample (e.g. pcolor plots for phi, theta).

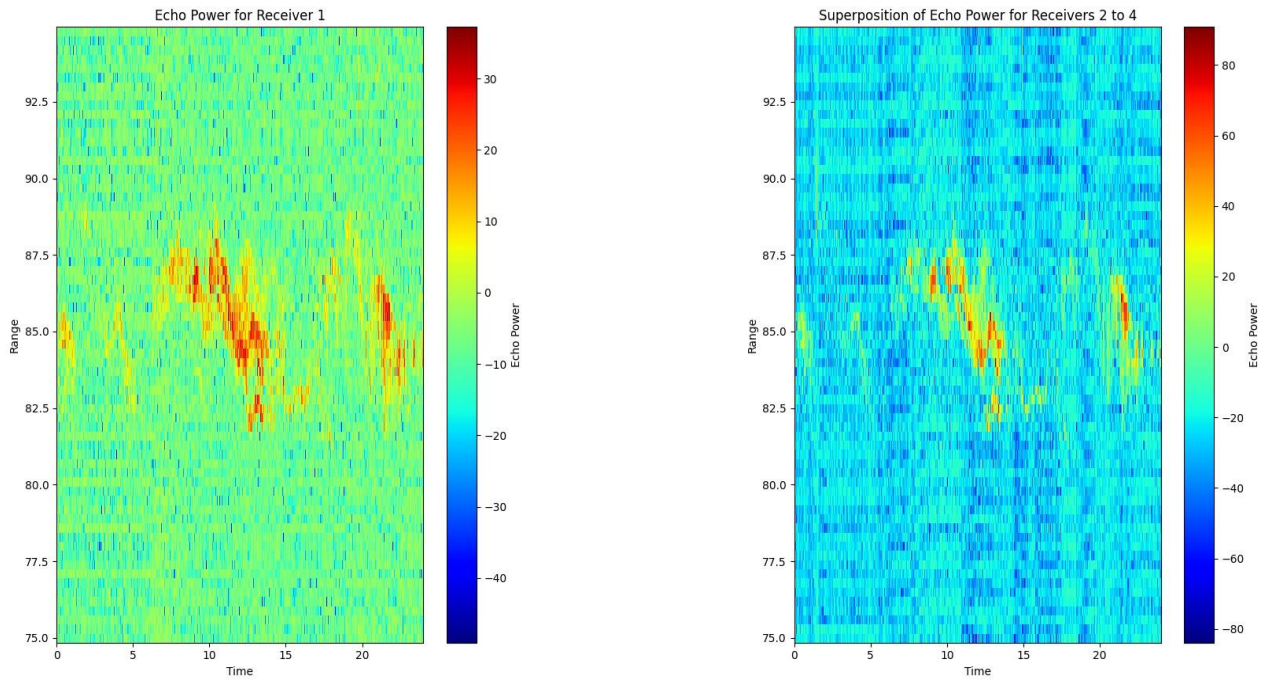


Fig 6.1 : Echo Power for receiver 1 and the superposition of power for receiver 2-4

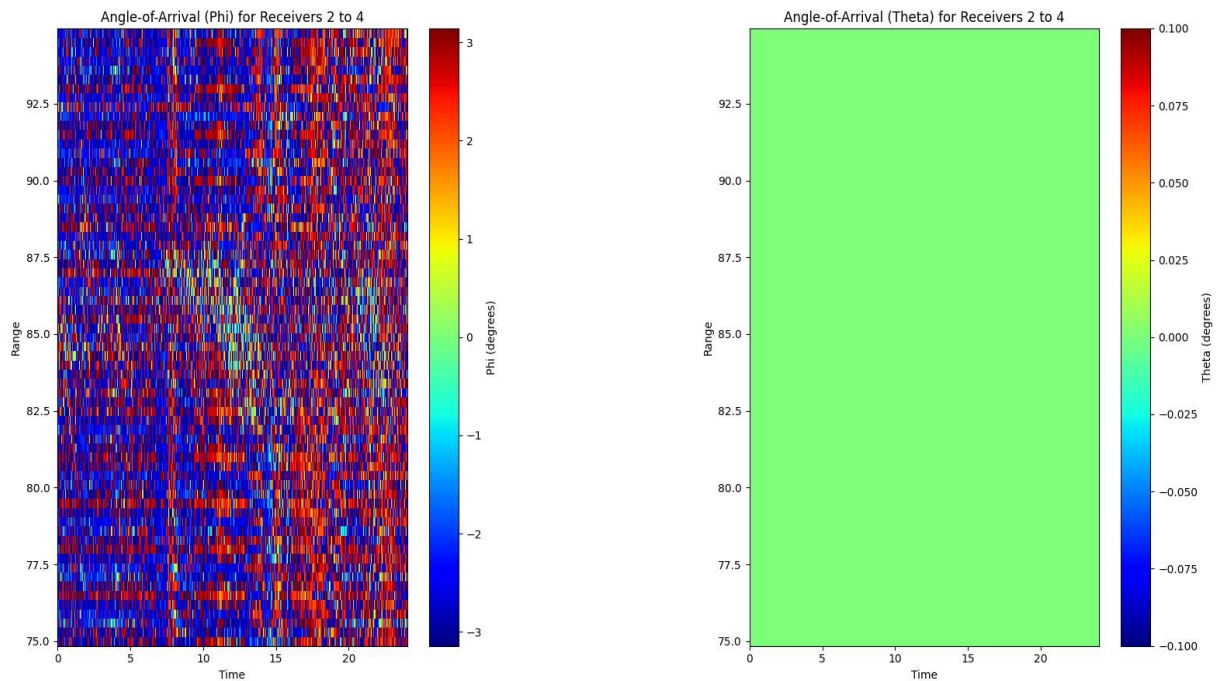


Fig 6.2 : AOA plot for receiver 2-4 for phi and theta

The echo power represents the energy or strength of the received echo signal and is usually measured in decibels (dB). It is directly related to the square of the amplitude of the received signal. A higher echo power indicates a stronger or more intense reflection, typically associated with larger or closer targets. In Figure 6.1, the plot illustrates the echo power for receiver 1 and the combined echo power for receivers 2 to 4. For receiver 1, the signal visibility is observed within a range of approximately 85 kilometers, with the maximum range reaching around 88 kilometers and the minimum range around 82 kilometers. The signal shows noticeable distortion, appearing discontinuous within certain time ranges. The superposition of echo power refers to the merging or summation of echo powers received from multiple receivers or sensor elements in a radar or sonar system. Each receiver independently captures its own echo signal, and by combining these individual powers, an aggregated echo power is obtained. In the case of receivers 2 to 4, the superimposed echo power exhibits a similar pattern, but the signal continuity is lower compared to receiver 1. The range covered by the superposition extends from approximately 87.5 kilometers (maximum range) to 82.5 kilometers (minimum range).

The Angle-of-Arrival (AoA) represents the direction from which a signal or wavefront arrives at a receiver or sensor. It is a crucial parameter used in various fields such as telecommunications, radar systems, wireless communication, and antenna array processing to determine the source, location, or direction of a signal. By analyzing the differences in signal arrival times or phases at multiple antennas or antenna elements, the AoA can be estimated. In this task, a 24-hour time-period dataset was collected from the available files. The superimposition of the receiver channels shows a similar pattern to the array data, albeit with higher noise levels. To calculate the AoA, the average phase was computed for each range gate and receiver pair. The resulting plots reveal a small variation between the theta and phi angles, indicating a vertical pointing beam direction. This averaging process helps reduce the variability in the angle estimation.

Task - 7

Interferometry - the shared RAW is NOT used HERE! -> SYNTHETIC !

Create your own sensor layout (circular (ring), spiral, random) in x-y-plane.

Use 6 sensors and test those for 2 different distributions (distances/spacing ~1 WL) between the sensors. First, assume equal phasing for all sensors and verify the vertical looking angle. (show corresponding plot) Simulate the coverage area (possible AOA-positions), by e.g. test MANY random phases for the sensors and derive the positions.

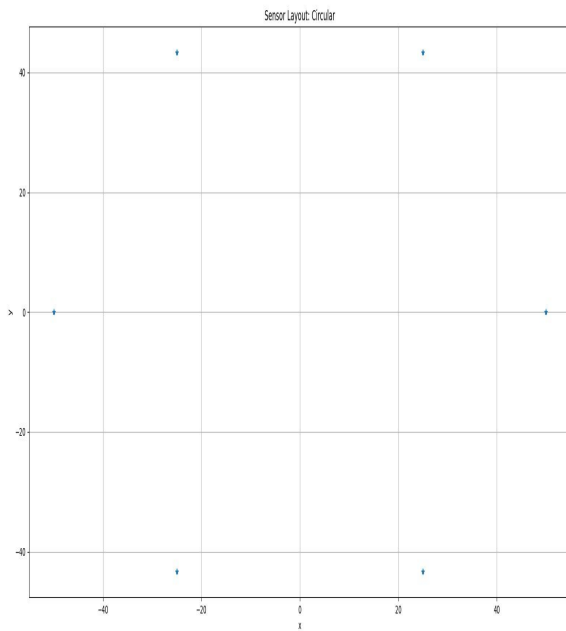


Fig 7.1 : Sensor Layout - Circular

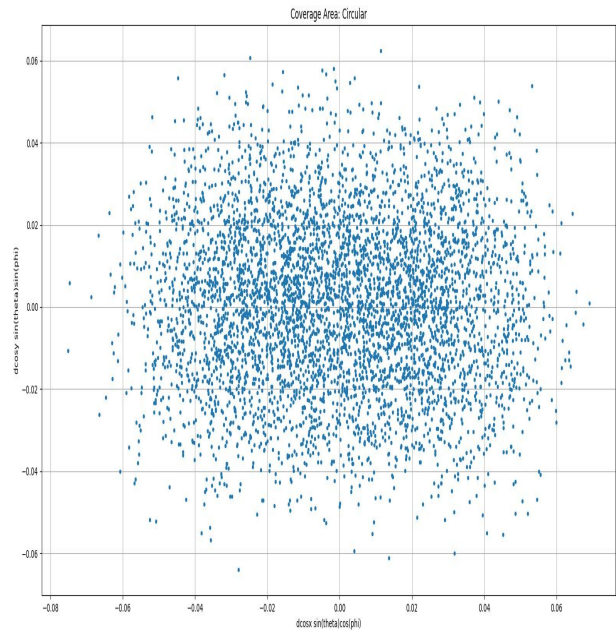


Fig 7.2 : Coverage Area - Circular

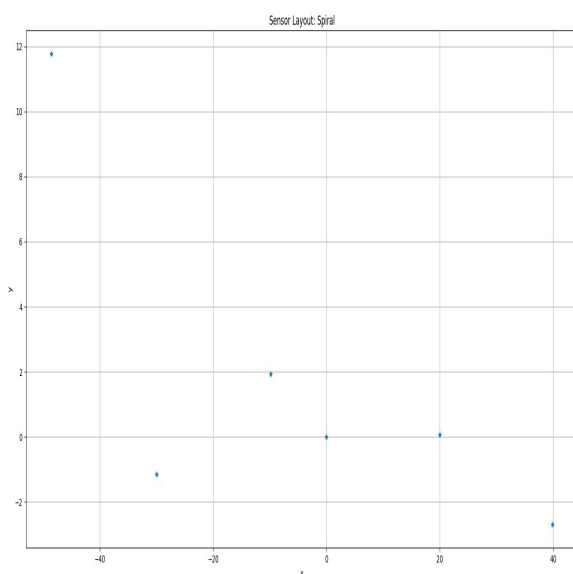


Fig 7.3 : Sensor Layout - Spiral

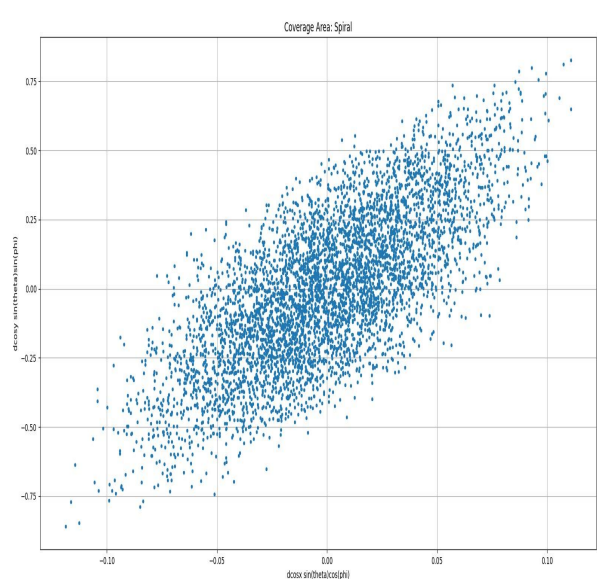


Fig 7.4 : Coverage Area - Spiral

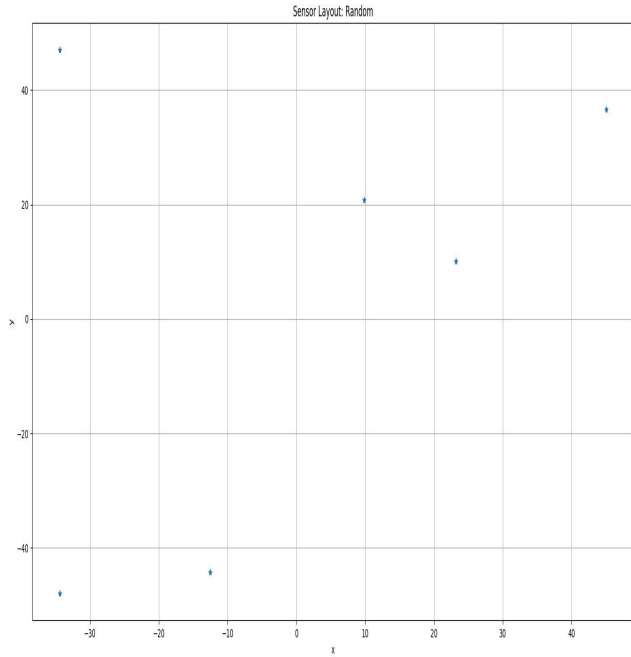


Fig 7.5 : Sensor Layout - Random

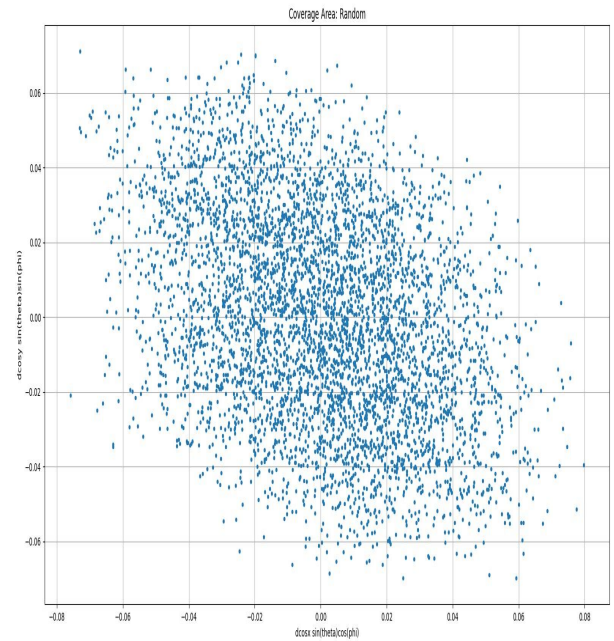


Fig 7.6 : Coverage Area - Random

Interferometry is a measurement technique that leverages wave interference to extract information about wave properties or objects that have interacted with the waves. It involves combining multiple coherent waves to create an interference pattern, which can be analyzed to gain valuable insights. Interferometry can be applied to various domains and utilizes electromagnetic waves (such as light or radio waves) or acoustic waves, depending on the specific application. The waves typically originate from a single source and are split into two or more paths before being recombined to create interference. The resulting interference pattern contains details about wave phase, amplitude, and coherence. Interferometry finds practical use in optics, astronomy, geodesy, radar, and telecommunications, among other fields. Figures 7.1, 7.3, and 7.5 depict circular, spiral, and random sensor layouts, respectively, where a configuration of six sensors is employed.

In the context of Angle-of-Arrival (AoA), the coverage area refers to the spatial region within which the positions of signal sources' AoA can be determined. The coverage area for AoA estimation depends on various factors, including the geometry of the antenna array, the aperture of the array, signal frequency, signal-to-noise ratio (SNR), propagation environment, and signal processing techniques. By carefully designing and optimizing these factors, it is possible to achieve a larger coverage area and accurate estimation of AoA for signal sources. Figures 7.2, 7.4, and 7.6 illustrate the coverage area plots corresponding to the circular, spiral, and random sensor layouts, respectively.

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